Pearson Edexcel Level 1/Level 2 – style practice paper

GCSE (9–1) Mathematics

1MA1/1H & 2H – Higher Tier

Topic 2: Algebra

Centre Number		**Candidate Number**	
Surname		**Other names**	

Practice questions in the style of Pearson Edexcel GCSE Mathematics (1MA1). Every question is original; the layout, command words, mark allocations and mark scheme (M, A, B marks) follow the real exam format. Calculator rules are given in the instructions below.

Time allowed	About 160 minutes (roughly 1 minute per mark)
Instructions	• Use **black** ink or ball-point pen. • Answer **all** questions. • Answer the questions in the spaces provided – there may be more space than you need. • Calculators must **not** be used in Section A. Calculators may be used in Section B. • Diagrams are **NOT** accurately drawn, unless otherwise indicated. • You must **show all your working**. • **Unless stated, give non-exact answers to 3 significant figures.**
Information	• The total mark for this paper is **158**. • The marks for **each** question are shown in brackets – use this as a guide as to how much time to spend on each question. • Questions labelled with an asterisk (*) are ones where the quality of your written communication will be assessed.
Advice	• Read each question carefully before you start to answer it. • Keep an eye on the time. • Try to answer every question. • Check your answers if you have time at the end.

Name: _____ Date: _____ Mark: _____ / 158

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Section A: Non-calculator

You must not use a calculator for questions 1 to 21.

1 (a) Expand and simplify $(x + 5)(x - 3)$ **(2)**

.....

(b) Expand $3x(2x^2 - 5)$ **(2)**

.....

(c) Expand and simplify $(2x + 3)(x - 4)(x + 1)$ **(3)**

.....

(Total for Question 1 is 7 marks)

2 (a) Factorise $x^2 - 7x - 18$ **(2)**

.....

(b) Factorise $6x^2 + x - 12$ **(3)**

.....

(c) Factorise fully $25y^2 - 16$ **(2)**

.....

(Total for Question 2 is 7 marks)

3 (a) Solve $5(2x - 3) = 3x + 13$ **(3)**

.....

(b) Solve $\frac{2x + 1}{3} - \frac{x - 4}{2} = 2$ **(4)**

.....

(Total for Question 3 is 7 marks)

4 (a) Make a the subject of $v^2 = u^2 + 2as$ **(2)**

.....

(b) Make x the subject of $y = \frac{3x + 2}{x - 1}$ **(4)**

.....

(Total for Question 4 is 6 marks)

5 (a) Solve the simultaneous equations $3x + 2y = 16$ and $x - y = 2$ **(3)**

.....

(b) Solve the simultaneous equations $x^2 + y^2 = 25$ and $y = x + 1$ **(5)**

.....

(Total for Question 5 is 8 marks)

6 (a) Solve $x^2 - 5x - 14 = 0$ **(3)**

.....

(b) Solve $2x^2 + 7x - 4 = 0$ **(3)**

.....

(c) Write $x^2 + 6x - 7$ in the form $(x + a)^2 + b$ where a and b are integers. **(2)**

.....

(d) Write down the coordinates of the turning point of the curve $y = x^2 + 6x - 7$. **(1)**

.....

(e) Show that the equation $2x^2 - 3x + 5 = 0$ has no real solutions. **(2)**

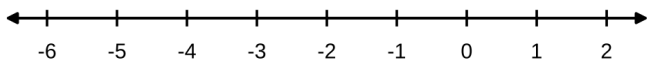
.....

(Total for Question 6 is 11 marks)

7 (a) Solve $5 - 2x \geq 11$ **(2)**

.....

(b) Show your answer to part (a) on the number line. **(1)**



(c) Solve $x^2 - 3x - 10 < 0$ **(3)**

.....

(Total for Question 7 is 6 marks)

8 Here are the first four terms of an arithmetic sequence. 7 11 15 19

(a) Find an expression for the nth term of the sequence. **(2)**

.....

(b) Is 203 a term in this sequence? You must show how you get your answer. **(2)**

.....

.....

(Total for Question 8 is 4 marks)

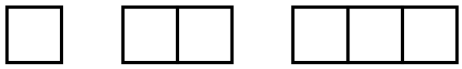
9 Here are the first four terms of a quadratic sequence. 3 9 19 33

Find an expression for the nth term. **(3)**

.....

(Total for Question 9 is 3 marks)

10 Here are the first three patterns in a sequence. Each pattern is made from matchsticks.



Pattern 1 Pattern 2 Pattern 3

(a) Draw Pattern 4. **(1)**

(b) Find an expression, in terms of n, for the number of matchsticks in Pattern n. **(2)**

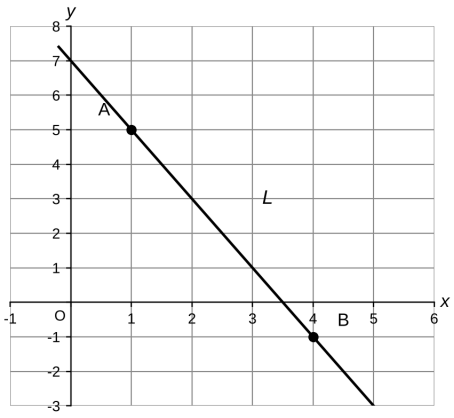
.....

(c) Dan uses 61 matchsticks to make a pattern. How many squares are in his pattern? **(2)**

.....

(Total for Question 10 is 5 marks)

11 The straight line L passes through the points $A(1, 5)$ and $B(4, -1)$.



(a) Find an equation of the line L. **(3)**

.....

(b) Write down the equation of the line that is parallel to $y = 3x - 2$ and passes through $(2, 9)$. **(2)**

.....

(c) Line M has equation $2y + x = 8$. Find an equation of the line that is perpendicular to M and passes through the point $(3, 1)$. **(4)**

.....

(Total for Question 11 is 9 marks)

12 (a) Show that $(n + 3)^2 - (n - 1)^2$ is a multiple of 8 for all positive integers n. **(3)**

.....

(b) Prove that the sum of two consecutive odd numbers is always a multiple of 4. **(3)**

.....

(Total for Question 12 is 6 marks)

13 $f(x) = 3x - 2$ and $g(x) = x^2 + 1$

(a) Work out $fg(2)$. **(2)**

.....

(b) Find $gf(x)$. Give your answer in the form $ax^2 + bx + c$. **(3)**

.....

(c) Find $f^{-1}(x)$. **(2)**

.....

(Total for Question 13 is 7 marks)

14 (a) Simplify fully $\frac{x^2 - 9}{x^2 + 5x + 6}$ **(3)**

.....

(b) Write $\frac{2}{x+1} + \frac{3}{x-2}$ as a single fraction in its simplest form. **(3)**

.....

(Total for Question 14 is 6 marks)

15 The diagram shows a rectangle. The lengths of the sides are shown. The perimeter of the rectangle is 40 cm.

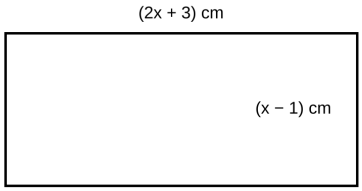


*Diagram **NOT** accurately drawn*

(a) Work out the value of x. **(3)**

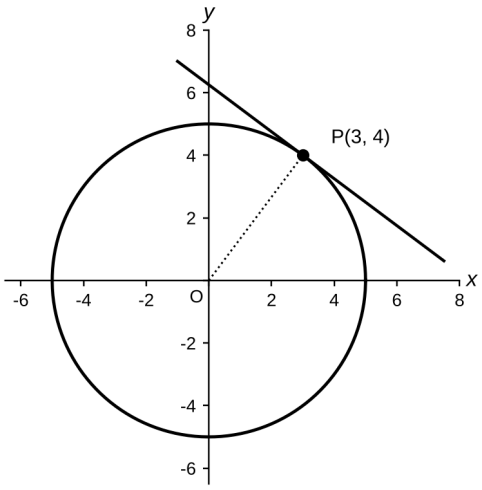
.....

(b) Work out the area of the rectangle. **(2)**

.....

(Total for Question 15 is 5 marks)

16 The circle $x^2 + y^2 = 25$ is drawn on the grid. The point $P(3, 4)$ lies on the circle. The straight line is a tangent to the circle at P.

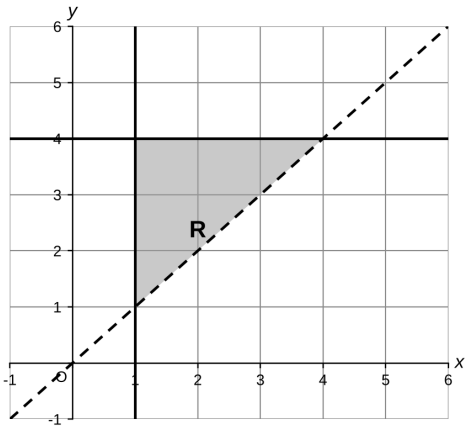


Find an equation of the tangent to the circle at P. **(4)**

.....

(Total for Question 16 is 4 marks)

17 The region R is shown shaded on the grid. The line $y = x$ is dashed.



Write down the three inequalities that define the region R. **(3)**

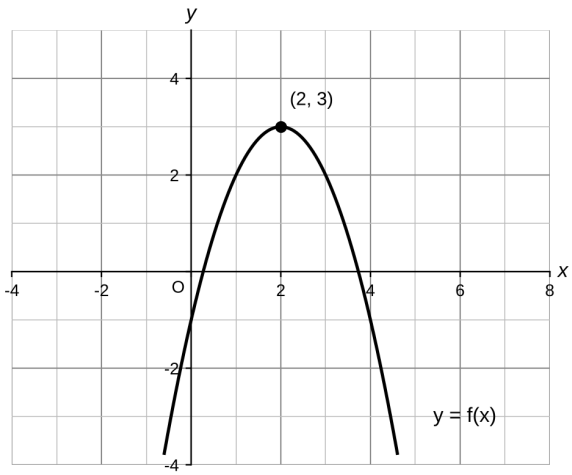
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(Total for Question 17 is 3 marks)

18 Here is the graph of $y = f(x)$. The turning point is $(2, 3)$.



(a) Write down the coordinates of the turning point of the graph of $y = f(x) + 2$. **(1)**

.....

(b) Write down the coordinates of the turning point of the graph of $y = f(x + 3)$. **(1)**

.....

(c) Write down the coordinates of the turning point of the graph of $y = -f(x)$. **(1)**

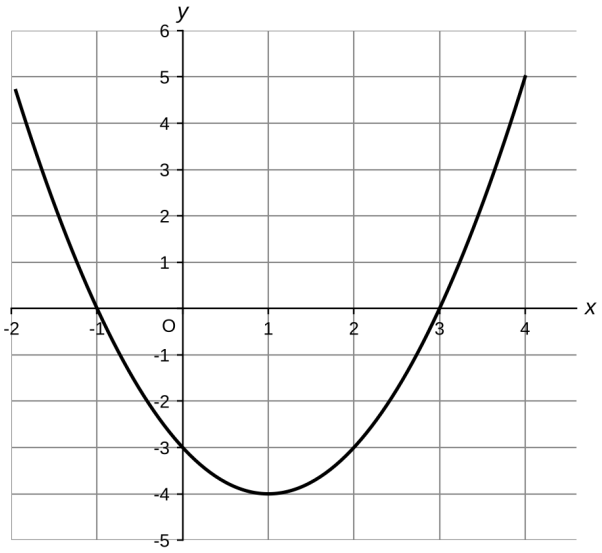
.....

(d) Write down the coordinates of the turning point of the graph of $y = f(2x)$. **(1)**

.....

(Total for Question 18 is 4 marks)

19 Here is the graph of $y = x^2 - 2x - 3$



(a) Write down the coordinates of the turning point of the curve. **(1)**

.....

(b) Write down the solutions of $x^2 - 2x - 3 = 0$ **(2)**

.....

(c) Use the graph to find estimates for the solutions of $x^2 - 2x - 3 = 2$ **(3)**

.....

(d) Write down the equation of the line of symmetry of the curve. **(1)**

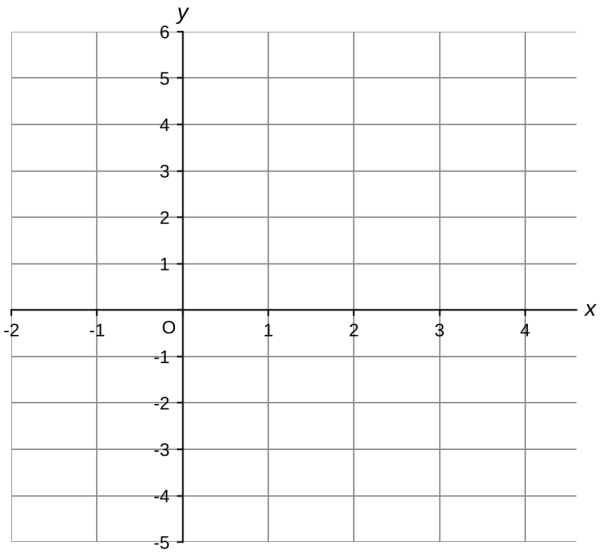
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(Total for Question 19 is 7 marks)

20 (a) Complete the table of values for $y = x^2 - 2x - 3$ **(2)**

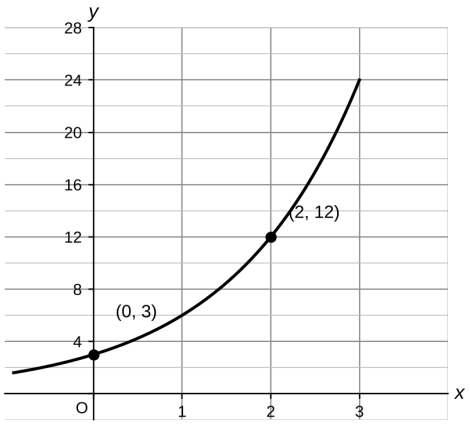
x	**−2**	**−1**	**0**	**1**	**2**	**3**	**4**
y							

(b) On the grid, draw the graph of $y = x^2 - 2x - 3$ for values of x from −2 to 4. **(2)**



(Total for Question 20 is 4 marks)

21 A curve has equation $y = ka^x$ where k and a are positive constants. The curve passes through the points $(0, 3)$ and $(2, 12)$.



Find the values of k and a. **(3)**

.....

(Total for Question 21 is 3 marks)

Section B: Calculator

A calculator may be used for questions 22 to 29.

22 Solve $3x^2 - 5x - 4 = 0$. Give your solutions correct to 3 significant figures. **(3)**

.....

(Total for Question 22 is 3 marks)

23 The equation $x^3 - 3x - 5 = 0$ has a solution between 2 and 3.

(a) Show that the equation $x^3 - 3x - 5 = 0$ can be rearranged to give $x = \sqrt[3]{3x + 5}$ **(2)**

.....

(b) Use the iteration formula $x_{n+1} = \sqrt[3]{3x_n + 5}$ with $x_0 = 2$ to find the values of x_1, x_2 and x_3. **(3)**

.....

(c) Show that the equation has a solution between 2 and 3 by evaluating $x^3 - 3x - 5$ at $x = 2$ and $x = 3$. **(2)**

.....

.....

.....

(Total for Question 23 is 7 marks)

24 The nth term of a sequence is $n^2 - 4n + 7$.

(a) Work out the 10th term of the sequence. **(2)**

.....

(b) Show that all the terms of the sequence are positive. **(3)**

.....

(Total for Question 24 is 5 marks)

25 The speed, v m/s, of a ball t seconds after it is kicked is given by $v = 20 - 5t$. Dev says, "The ball stops after 5 seconds."

Is Dev correct? You must show how you get your answer. **(2)**

.....

.....

.....

(Total for Question 25 is 2 marks)

26 A rectangular field has a perimeter of 120 m. The length of the field is x metres.

(a) Show that the area, A m², of the field is given by $A = 60x - x^2$. **(2)**

.....

(b) Find the value of x that gives the greatest possible area, and work out this area. **(3)**

.....

(Total for Question 26 is 5 marks)

27 Sam is x years old. Her brother is 4 years older than her. The product of their ages is 221.

Work out how old Sam is. **(4)**

.....

(Total for Question 27 is 4 marks)

28 Solve algebraically the simultaneous equations $y = 2x^2 - 3x + 1$ and $y = x + 7$. Give your answers correct to 2 decimal places where necessary.

(5)

.....

(Total for Question 28 is 5 marks)

29 A bank pays compound interest. After n years, the amount in an account is $A = 2500 \times 1.04^n$.

(a) Work out the amount in the account after 8 years. **(2)**

£ .....

(b) Find the least number of whole years for the amount to be more than £5000. **(3)**

.....

(Total for Question 29 is 5 marks)

TOTAL FOR PAPER IS 158 MARKS

END

MARK SCHEME – Topic 2: Algebra

Guidance on marking: M marks are method marks, awarded for a correct method (even if the final answer is wrong). **A** marks are accuracy marks, awarded only if the relevant M mark has been gained. **B** marks are independent marks (no method needed). **C** marks are communication marks. **P** marks are process marks (a correct approach to a problem). **ft** = follow through from a previous incorrect answer. **oe** = or equivalent. **cao** = correct answer only. **SC** = special case. If the answer on the answer line differs from the working, the answer line is marked unless the working is clearly the intended answer. Correct answers with no working score full marks unless the question says 'You must show your working'.

Question	Working	Answer	Mark	Notes
1(a)	M1 $x^2 - 3x + 5x - 15$ (3 of 4 terms correct) A1 $x^2 + 2x - 15$	$x^2 + 2x - 15$	2 AO1	
1(b)	M1 one term correct A1 $6x^3 - 15x$	$6x^3 - 15x$	2 AO1	
1(c)	M1 $(2x + 3)(x - 4) = 2x^2 - 5x - 12$ M1 $(2x^2 - 5x - 12)(x + 1)$ with at least 4 terms correct A1 $2x^3 - 3x^2 - 17x - 12$	$2x^3 - 3x^2 - 17x - 12$	3 AO1	
2(a)	M1 $(x \pm 9)(x \pm 2)$ A1 $(x - 9)(x + 2)$	$(x - 9)(x + 2)$	2 AO1	
2(b)	M1 $(3x \pm \ldots)(2x \pm \ldots)$ or $6x^2 + 9x - 8x - 12$ M1 $3x(2x + 3) - 4(2x + 3)$ A1 $(3x - 4)(2x + 3)$	$(3x - 4)(2x + 3)$	3 AO1	
2(c)	M1 $(5y \pm 4)(5y \pm 4)$ A1 $(5y - 4)(5y + 4)$	$(5y - 4)(5y + 4)$	2 AO1	
3(a)	M1 $10x - 15$ M1 $10x - 3x = 13 + 15$ A1 $x = 4$	$x = 4$	3 AO1	
3(b)	M1 multiply by 6: $2(2x + 1) - 3(x - 4) = 12$ M1 $4x + 2 - 3x + 12 = 12$ M1 $x + 14 = 12$ A1 $x = -2$	$x = -2$	4 AO1	
4(a)	M1 $v^2 - u^2 = 2as$ A1 $a = \dfrac{v^2 - u^2}{2s}$	$a = \dfrac{v^2 - u^2}{2s}$	2 AO1	
4(b)	M1 $y(x - 1) = 3x + 2$ M1 $yx - y = 3x + 2$ M1 $yx - 3x = y + 2$ and $x(y - 3) = y + 2$ A1 $x = \dfrac{y + 2}{y - 3}$	$x = \dfrac{y + 2}{y - 3}$	4 AO2	
5(a)	M1 correct method to eliminate one variable (e.g. $3x + 2y = 16$ and $2x - 2y = 4$) M1 (dep) substitute to find the other variable A1 $x = 4$, $y = 2$	x = 4, y = 2	3 AO1	
5(b)	M1 $x^2 + (x + 1)^2 = 25$ M1 $2x^2 + 2x - 24 = 0$ (or $x^2 + x - 12 = 0$) M1 $(x + 4)(x - 3) = 0$ A1 $x = -4$ and $x = 3$ A1 $(-4, -3)$ and $(3, 4)$	(−4, −3) and (3, 4)	5 AO3	

Question	Working	Answer	Mark	Notes
6(a)	M1 $(x-7)(x+2)$ A1 $x=7$ A1 $x=-2$	x = 7 or x = −2	3 AO1	
6(b)	M1 $(2x-1)(x+4)$ A1 $x=\frac{1}{2}$ A1 $x=-4$	x = ½ or x = −4	3 AO1	
6(c)	M1 $(x+3)^2$ A1 $(x+3)^2-16$	$(x+3)^2-16$	2 AO1	
6(d)	B1 $(-3,\ -16)$ (ft from part (c))	(−3, −16)	1 AO1	
6(e)	M1 $b^2-4ac=(-3)^2-4\times2\times5$ C1 $=-31<0$ so no real solutions		2 AO2	
7(a)	M1 $-2x\geq6$ or $5-11\geq2x$ A1 $x\leq-3$	x ≤ −3	2 AO1	
7(b)	B1 closed (solid) circle at -3 with arrow towards the left		1 AO1	
7(c)	M1 $(x-5)(x+2)$ or critical values 5 and -2 M1 sketch or sign test showing the region between the roots A1 $-2<x<5$	−2 < x < 5	3 AO1	
8(a)	M1 $4n$ seen A1 $4n+3$	$4n+3$	2 AO1	
8(b)	M1 $4n+3=203$ C1 $n=50$ (a whole number) so yes, it is the 50th term		2 AO2	
9	M1 first differences 6, 10, 14 and second difference 4 M1 $2n^2$ seen and subtract: 1, 1, 1, 1 A1 $2n^2+1$	$2n^2+1$	3 AO1	
10(a)	B1 four squares in a row (13 matchsticks)		1 AO1	
10(b)	M1 4, 7, 10 or common difference 3 A1 $3n+1$	$3n+1$	2 AO2	
10(c)	M1 $3n+1=61$ A1 20	20	2 AO3	
11(a)	M1 gradient $\frac{-1-5}{4-1}$ A1 -2 A1 $y=-2x+7$	y = −2x + 7	3 AO2	
11(b)	M1 $y=3x+c$ with $9=6+c$ A1 $y=3x+3$	y = 3x + 3	2 AO1	
11(c)	M1 rearrange: $y=-\frac{1}{2}x+4$, gradient $-\frac{1}{2}$ M1 perpendicular gradient 2 M1 $y-1=2(x-3)$ or $1=6+c$ A1 $y=2x-5$	y = 2x − 5	4 AO3	
12(a)	M1 $(n^2+6n+9)-(n^2-2n+1)$ M1 $8n+8$ C1 $=8(n+1)$, which is a multiple of 8		3 AO2	

Question	Working	Answer	Mark	Notes
12(b)	M1 $2n+1$ and $2n+3$ (or $2n-1$ and $2n+1$) M1 $4n+4$ C1 $=4(n+1)$ so always a multiple of 4		3 AO2	
13(a)	M1 $g(2)=5$ A1 13	13	2 AO1	
13(b)	M1 $(3x-2)^2+1$ M1 $9x^2-6x-6x+4+1$ A1 $9x^2-12x+5$	$9x^2-12x+5$	3 AO1	
13(c)	M1 $y=3x-2 \rightarrow x=\dfrac{y+2}{3}$ A1 $f^{-1}(x)=\dfrac{x+2}{3}$	$f^{-1}(x)=\dfrac{x+2}{3}$	2 AO1	
14(a)	M1 $(x-3)(x+3)$ M1 $(x+2)(x+3)$ A1 $\dfrac{x-3}{x+2}$	$\dfrac{x-3}{x+2}$	3 AO1	
14(b)	M1 common denominator $(x+1)(x-2)$ M1 $2(x-2)+3(x+1)$ A1 $\dfrac{5x-1}{(x+1)(x-2)}$	$\dfrac{5x-1}{(x+1)(x-2)}$	3 AO1	
15(a)	M1 $2(2x+3)+2(x-1)=40$ M1 $6x+4=40$ A1 $x=6$	x = 6	3 AO3	
15(b)	M1 $(2\times 6+3)\times(6-1)$ A1 75	75 cm²	2 AO3	
16	M1 gradient of $OP=\dfrac{4}{3}$ M1 gradient of tangent $=-\dfrac{3}{4}$ M1 $y-4=-\dfrac{3}{4}(x-3)$ A1 $3x+4y=25$ (oe)	3x + 4y = 25	4 AO3	
17	B1 $x\geq 1$ B1 $y\leq 4$ B1 $y>x$	x ≥ 1, y ≤ 4, y > x	3 AO2	
18(a)	B1 $(2, 5)$	(2, 5)	1 AO1	
18(b)	B1 $(-1, 3)$	(−1, 3)	1 AO1	
18(c)	B1 $(2, -3)$	(2, −3)	1 AO1	
18(d)	B1 $(1, 3)$	(1, 3)	1 AO1	
19(a)	B1 $(1, -4)$	(1, −4)	1 AO1	
19(b)	B1 $x=-1$ B1 $x=3$	x = ... or x = ...	2 AO1	

Question	Working	Answer	Mark	Notes
19(c)	M1 draws the line $y = 2$ (or reads the curve at $y = 2$) A1 $-1.5 \leq x \leq -1.4$ A1 $3.4 \leq x \leq 3.5$	x = ... or x = ...	3 AO3	
19(d)	B1 $x = 1$	x = 1	1 AO1	
20(a)	B2 all correct: $5,\ 0,\ -3,\ -4,\ -3,\ 0,\ 5$ for $x = -2,\ -1,\ 0,\ 1,\ 2,\ 3,\ 4$ (B1 for any 4 correct)		2 AO1	
20(b)	M1 at least 5 points plotted correctly (ft) A1 smooth curve through all points		2 AO1	
21	B1 $k = 3$ (from $y = 3$ when $x = 0$) M1 $12 = 3a^2$ A1 $a = 2$	k = 3, a = 2	3 AO3	
22	M1 $\frac{-(-5) \pm \sqrt{(-5)^2 - 4 \times 3 \times (-4)}}{2 \times 3}$ (allow one slip) M1 $\frac{5 \pm \sqrt{73}}{6}$ A1 2.26 and -0.591	x = ... or x = ...	3 AO1	
23(a)	M1 $x^3 = 3x + 5$ C1 take the cube root of both sides		2 AO2	
23(b)	B1 $x_1 = 2.22398\ldots$ B1 $x_2 = 2.26837\ldots$ B1 $x_3 = 2.27696\ldots$	x_1 = ..., x_2 = ..., x_3 = ...	3 AO1	
23(c)	M1 $f(2) = -3$ and $f(3) = 13$ C1 change of sign, so a root lies between 2 and 3 (continuous function)		2 AO2	
24(a)	M1 $10^2 - 40 + 7$ A1 67	67	2 AO1	
24(b)	M1 complete the square: $(n-2)^2 + 3$ M1 a square is ≥ 0 so the expression is ≥ 3 C1 hence always positive		3 AO2	
25	M1 $20 - 5t = 0$ or $v = 20 - 25 = -5$ at $t = 5$ C1 $t = 4$, so Dev is not correct		2 AO2	
26(a)	M1 width $= 60 - x$ C1 $A = x(60 - x) = 60x - x^2$		2 AO2	
26(b)	M1 completing the square: $A = 900 - (x-30)^2$ or symmetry A1 $x = 30$ A1 900 m²	x = 30, area = 900 m²	3 AO3	
27	M1 $x(x+4) = 221$ M1 $x^2 + 4x - 221 = 0$ M1 $(x+17)(x-13) = 0$ or formula A1 13 (reject -17)	13 years	4 AO3	

Question	Working	Answer	Mark	Notes
28	M1 $2x^2 - 3x + 1 = x + 7$ M1 $2x^2 - 4x - 6 = 0 \rightarrow x^2 - 2x - 3 = 0$ M1 $(x - 3)(x + 1) = 0$ A1 $x = 3$ and $x = -1$ A1 $(3, 10)$ and $(-1, 6)$	(3, 10) and (−1, 6)	5 AO3	
29(a)	M1 2500×1.04^8 A1 3421.42 (accept 3421.4…)	£ 3421.42	2 AO1	
29(b)	M1 $2500 \times 1.04^n > 5000$ or trial of values M1 $1.04^{17} = 1.9479\ldots$ and $1.04^{18} = 2.0258\ldots$ A1 18	18 years	3 AO3	

Where marks are lost in Topic 2

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Expanding brackets:** in a double bracket write all four products before collecting. Watch the sign of the −3x and +5x terms.
- ☐ **Factorising:** always multiply back out to check. For $ax^2 + bx + c$ find two numbers that multiply to ac and add to b.
- ☐ **Changing the subject:** if the subject appears twice, collect those terms on one side and factorise.
- ☐ **Simultaneous (linear + circle/quadratic):** substitute the linear equation into the other, then *go back* and find both y values (or x values). Write the answers as pairs.
- ☐ **Quadratic formula:** brackets round $-b$ and b^2 on the calculator. Give the answer to the accuracy asked.
- ☐ **Inequalities:** dividing or multiplying by a negative flips the sign. For $x^2 < k$ type inequalities give a *double* inequality.
- ☐ **nth term:** first differences constant → linear; second differences constant → quadratic ($\frac{1}{2}$ of the second difference × n^2).
- ☐ **Proof:** start with algebra for general numbers ($2n$, $2n + 1$), show the result is a multiple of the target and *say so* in a sentence.
- ☐ **Graph transformations:** $f(x + a)$ moves left by a; $f(ax)$ stretches horizontally by $\frac{1}{a}$.